We've walked through various Databricks products and features geared toward helping you understand your data. Get ready to follow along in your own Databricks workspace as we work through the *Chapter 4* code by project and put these concepts into practice.

Applying our learning

It's time to apply these concepts to our example projects. We will use what we have learned to explore each project dataset, from using Databricks Assistant to AutoML, to creating a vector search index and exploring image data.

Technical requirements

Before you begin, review, and prepare the technical requirements necessary for the hands-on work in this chapter:

- We use the `missingno` library to address missing numbers in our synthetic transactions project data: `https://pypi.org/project/missingno/`
- For the RAG project, you will need to install the following either on your cluster or in the `CH4-01-Creating_VectorDB` notebook. If you choose to install them in the notebook, the code is included for you:
  - `typing_extensions==4.7.1`
  - `transformers==4.30.2`
  - `llama-index==0.9.3`
  - `langchain==0.0.319`
  - `unstructured[pdf,docx]==0.10.30`

Project: Favorita Store Sales – Time Series Forecasting

For the Favorita Store Sales project, we use many simple DBSQL queries for data exploration and to understand the relationships between datasets. Additionally, we use the `ydata_profiling` library to produce data profiles in HTML format, as shown in *Figure 4.19*. To follow along in your own workspace, please refer to the following notebooks:

- `CH4-01-Exploring Favorita Sales Data`
- `CH4-02-Exploring Autogenerated Notebook`
- `CH4-03-Imputing Oil Data`